The Cabibbo Angle from Geometry

Quark Mixing Emerges from String Dimensions

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Abstract

The Cabibbo angle theta_c governs quark mixing between generations. Its measured value sin(theta_c) = 0.2253 has no explanation in the Standard Model. We derive **sin(theta_c) = 1/sqrt(20) = 0.2236** from Z^2 geometry, achieving **0.75% accuracy**. The factor 20 = 2 × 10 comes from two-generation mixing (2) times superstring dimensions (10 = GAUGE - 2). We also derive the Jarlskog invariant J = 1/(1000 × Z^2) with 3.1% accuracy, explaining CP violation geometrically.

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1. The Problem

1.1 The Cabibbo Angle

The weak force allows transitions between quark generations. This is described by the CKM matrix, with the largest off-diagonal element being:

|V_us| = sin(theta_c) = 0.2253 ± 0.0008

This is the Cabibbo angle, measured to high precision.

1.2 Why Is This a Problem?

In the Standard Model, the Cabibbo angle is a free parameter. There is no principle that determines its value. Why is sin(theta_c) approximately 0.22 rather than some other number?

1.3 Historical Attempts

- Fritzsch-Xing relations: Empirical patterns without derivation
- Texture zeros: Ad hoc assumptions about mass matrices
- Family symmetries: Many models, no unique prediction

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2. The Z^2 Solution

2.1 The Formula

$\sin^2(\text{theta_c}) = 1/(2 \times (\text{GAUGE} - 2))$

$\sin^2(\text{theta_c}) = 1/(2 \times 10) = 1/20$

$\sin(\text{theta_c}) = 1/\text{sqrt}(20) = 0.2236$

2.2 The Components

Component	Value	Meaning
GAUGE	12	Standard Model gauge generators
GAUGE - 2	10	Superstring dimensions
2	2	Two generations mixing
Result	**1/20**	**$\sin^2(\text{theta_c})$**

2.3 Verification

Quantity	Predicted	Measured	Error
sin(theta_c)	0.2236	0.2253	0.75%
theta_c	12.92°	13.02°	0.77%

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3. Why 1/sqrt(20)?

3.1 The Structure

$20 = 2 \times 10$

where: - **2** = number of generations participating in the mixing - **10 = GAUGE - 2** = superstring critical dimension

3.2 Physical Meaning

Factor of 2: - The Cabibbo angle describes 1-2 generation mixing - Specifically: u-d ? c-s transitions - Two generations participate

Factor of 10: - From Z^2: GAUGE = 12 - String dimensions: D = GAUGE - 2 = 10 - This is the superstring critical dimension - Mixing is suppressed by string geometry

3.3 The Deep Connection

$\sin^2(\text{theta_c}) = 1/(\text{n_generations} \times \text{D_string})$

where: - n_generations = 2 (for Cabibbo mixing) - D_string = 10 (superstring dimensions)

The Cabibbo angle “knows about” extra dimensions!

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4. The Jarlskog Invariant

4.1 CP Violation Measure

The Jarlskog invariant quantifies CP violation in quark mixing:

J = Im(V_us V_cb V*_ub V*_cs)

Measured: $J = (3.08 \pm 0.13) \times 10^{-5}$

4.2 Z^2 Derivation

$J = 1/(1000 \times Z^2)$

$J = 1/(1000 \times 33.51) = 2.98 \times 10^{-5}$

4.3 Verification

Quantity	Predicted	Measured	Error
J	2.98×10^{-5}	3.08×10^{-5}	3.1%

4.4 Physical Meaning

$J = 1/(10^3 \times Z^2)$

- Factor 10^3 = D_string × 100: String dimension times century
- Factor Z^2: Geometric normalization
- CP violation is geometrically determined

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5. The CKM Hierarchy

5.1 Wolfenstein Parameterization

The CKM matrix has hierarchical structure:

|V_us| ~ lambda = sin(theta_c) ~ 0.22 |V_cb| ~ lambda² ~ 0.04 |V_ub| ~ lambda³ ~ 0.004

5.2 Z^2 Explanation

lambda = sin(theta_c) = 1/sqrt(20)

The hierarchy follows powers of lambda:

Element	Scaling	Z^2 Value
V_us	lambda	0.224
V_cb	lambda²	0.050
V_ub	lambda³	0.011

The CKM hierarchy is a geometric progression from the fundamental scale 1/sqrt(20).

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6. Connection to Other Z^2 Results

6.1 The Web

The Cabibbo angle connects to:

1. String dimensions: D = GAUGE - 2 = 10 (appears in denominator)

2. Fine structure constant: alpha^{-1} = $4Z^2 + 3$ = 137.04 (related suppressions)

3. Three generations: N_gen = 3 (full CKM is 3×3 matrix)

4. Neutrino mixing: Different pattern (tribimaximal vs hierarchical) But both from Z^2

6.2 Quark vs Lepton Mixing

Property	Quarks (CKM)	Leptons (PMNS)
theta_12	13° (Cabibbo)	34° (solar)
Pattern	Hierarchical	Democratic
Formula	1/sqrt(2D)	1/3

Quark mixing is suppressed by string dimensions. Lepton mixing reflects generational democracy. Both derive from Z^2.

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7. Predictions

7.1 The Cabibbo Angle Is Exact

Z^2 predicts: sin(theta_c) = 1/sqrt(20) = 0.22360679…

Future precision should confirm this exact value.

7.2 The Hierarchy Persists

The lambda expansion (powers of sin(theta_c)) is fundamental, not accidental.

7.3 CP Violation Is Fixed

$J = 1/(1000 \times Z^2) = 2.98 \times 10^{-5}$

CP violation in quarks is geometrically determined.

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8. Summary

The Problem

Why is sin(theta_c) ≈ 0.22?

The Standard Model

Cannot explain it. It's a free parameter.

The Z^2 Answer

sin(theta_c) = 1/sqrt(2 × (GAUGE - 2)) = 1/sqrt(20)

Term	Value	Meaning
GAUGE - 2	10	String dimensions
2	2	Generation pair
sin(theta_c)	**0.2236**	**0.75% from measured**

The Significance

Quark mixing is determined by: - String geometry (D = 10) - Generation structure (2 generations mixing)

Both derive from Z^2 = CUBE × SPHERE = 32pi/3.

Appendix: The Calculation

A.1 Z^2 Constant

$Z^2 = 8 \times (4pi/3) = 32pi/3 = 33.5103$

A.2 GAUGE

$GAUGE = 9Z^2/(8pi) = 12$

A.3 String Dimensions

$D = GAUGE - 2 = 10$

A.4 Cabibbo Angle

$\sin^2(theta_c) = 1/(2D) = 1/20$ $\sin(theta_c) = 0.22361$ $theta_c = 12.92°$

A.5 Comparison

Measured: $\sin(theta_c) = 0.2253$ Error: $|0.2236 - 0.2253|/0.2253 = 0.75\%$

"Quark mixing comes from string dimensions through Z^2."

— Carl Zimmerman, 2026

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Repository: https://github.com/carlzimmerman/zimmerman-formula